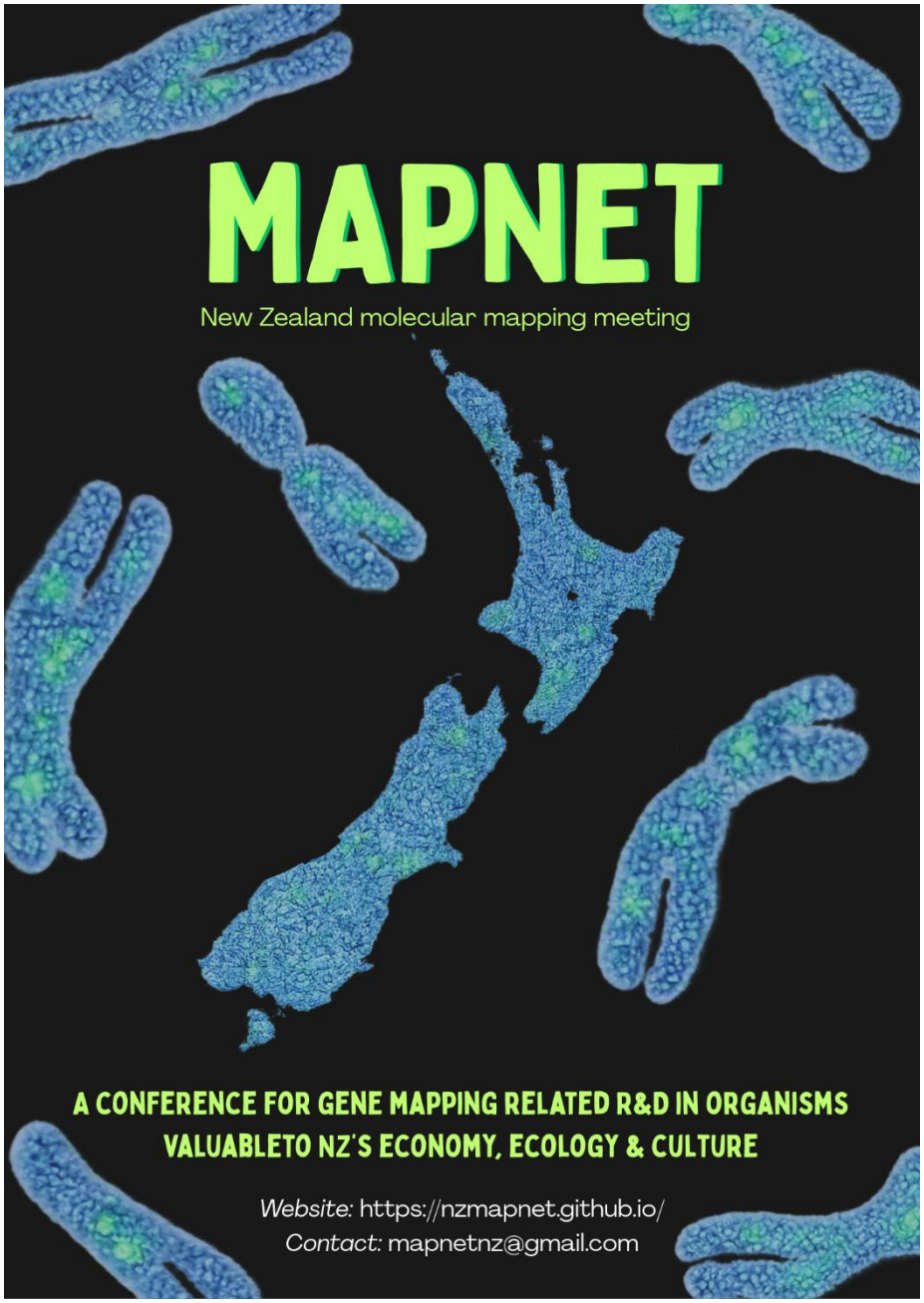
MapNet 2026

Programme & Abstract Book

MapNet Meeting, Kirikiriroa – Hamilton



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Mihimihi

E ngā mana, e ngā reo, e ngā karangatanga maha, tēnā koutou! Nau mai, haere mai ki te hui nei o MapNet mō te tau 2026. Ko te hui nei ki Newstead, Kirikiriroa, ā, ko LIC te kaiwhakahaere manaaki. Te āhua o te hui nei kei roto i te whakataukī: Mā te whakarongo ka mōhio, mā te mōhio ka mārama, mā te mārama ka mātau, mā te mātau ka ora. Nō reira, e ngā kairangahau kua tae mai, nau mai, mihi mai, karanga mai ki a koutou katoa.

Translation: All peoples, all languages, all authorities, greetings! Welcome to the MapNet 2026 conference, held in Newstead, Hamilton, and hosted by LIC. The theme for the conference is embodied in the whakataukī (proverb) 'Mā te whakarongo ka mōhio, mā te mōhio ka mārama, mā te mārama ka mātau, mā te mātau ka ora.' This translates to: 'Through listening comes knowledge, through knowledge comes understanding, through understanding comes wisdom, through wisdom comes wellbeing.' So, to all researchers gathered here: greetings and welcome to you all.

About MapNet:

Established in 2005, MapNet is a semi-formal Aotearoa New Zealand-wide collective of researchers working across gene mapping, quantitative genetics, genomics, and genetical genomics. Its purpose is to encourage pre-commercial, cross-sector research and development that supports gene mapping and genomics in organisms of value to Aotearoa New Zealand's economy, ecology, health, and culture.

Participation in MapNet is voluntary and centred on individual researchers, with representation across health, environment, primary industries, and emerging areas of genomic research. MapNet activities include annual workshops, collaborative research initiatives, one-off community events, and advocacy on behalf of the Aotearoa gene mapping and genomics research community.

MapNet is overseen by a steering committee and a convenor.

Current members are:

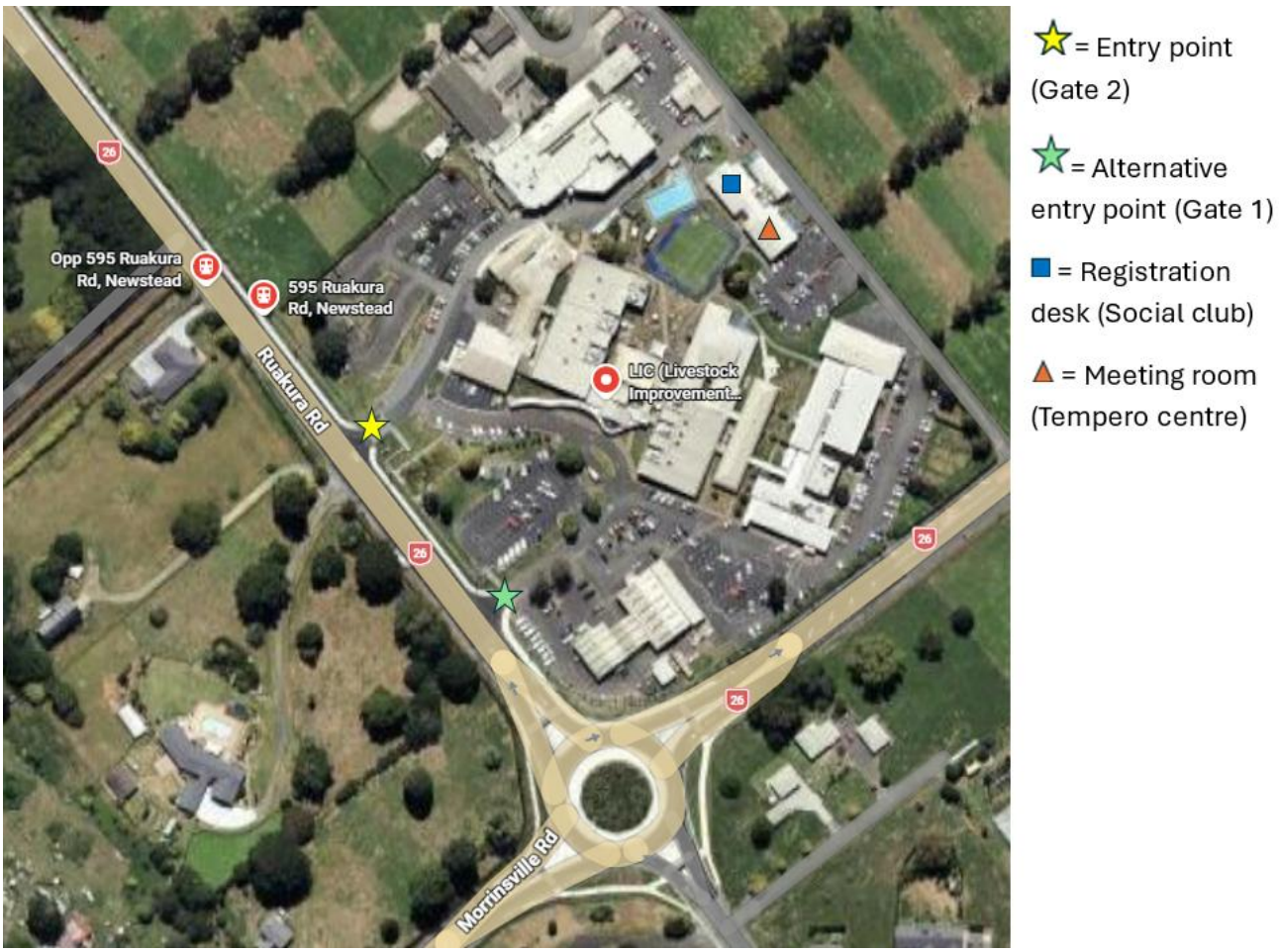
Dr Natalie Graham
Dr Annabel Whibley
Dr Alastair Lamont
Dr Rachel Ashby
Dr John McCallum
Dr Roy Castilla
Professor Phillip Wilcox (Convenor)

Organising Committee members for MapNet 2026

Dr Laura Duntsch (LIC)
Dr Emma Voss (LIC)
Dr Fenella Deans (LIC)
Hannah Trebes (LIC)
Dr Thomas Lopdell (LIC)
Dr Chad Harland (LIC)
Dr Natalie Graham (Bioeconomy Science Institute)
Dr Steffi Fritsche (Bioeconomy Science Institute)

Location

Conference sessions will be held at the LIC campus in Newstead - 605 Ruakura Road Newstead 3286, in the Tempero Centre.



Social Function

Conference dinner is on Tuesday evening from 6:30pm at Zenders - 44 Hopkins Road, Newstead Hamilton 3286.

Photography

Please note that a member of the LIC team will be taking photographs throughout the event. These photos may be used in LIC communications, including on LIC's LinkedIn page and other promotional materials. If you do not wish to have your photo taken or shared online, please let a member of the organising team know before or during the event.

MapNet Website

https://nzmapnet.github.io/

Code of conduct

We are dedicated to providing a harassment-free meeting for everyone. All attendees, speakers, and sponsors are required to abide by the code of conduct located on the MapNet website. Organisers will enforce the code of conduct and expect cooperation to ensure a safe environment for all.

Please respect the requests of speakers and conference attendees that ask or suggest not to be included in social media posts.

Biosecurity notice

Applies to all visitors to LIC offices and/or farms

The following regulations apply to ensure LIC maintains its Quarantine status – Visitors who have had contact with an overseas bovine disease outbreak, exotic to New Zealand will not be admitted to LIC sites unless:

* They have been passed by the New Zealand Biosecurity Regulation inspections at point of entry to New Zealand and carry no materials/footwear that could potentially put LIC's operations at risk from infection from bovine disease outbreak.

All potential visitors to LIC sites in New Zealand who, within 30 days prior to their visit/first entry to the sites, have been in or near the vicinity of a bovine disease outbreak (exotic to New Zealand) overseas, MUST:

* Complete a Quarantine Declaration (available on request) detailing their movements overseas.

Tuesday 3 November			
08:30–09:00	Registration & coffee		
09:00–09:15	Opening Karakia: Phillip Wilcox Housekeeping		
09:15–10:15	**Chair: Emma Voss** **Keynote Speaker:** Maren Wellenreuther - Breeding for an Uncertain Future		
10:15–10:45	**Morning Tea**		
10:45–12:00	**Session 1: Aquaculture, Conservation and Companion Animals** **Chair: Emma Voss**		
	10:45-11:00	Genomic erosion in the threatened hihi / stitchbird	Anna Santure
	11:00-11:15	Genomic origins and population structures of the Huntaway and Heading Dog breeds in New Zealand	Milly Henry
	11:15-11:30	Structural variant size, type and genomic context shape continuous growth variation in snapper (*Chrysophrys auratus*)	Julie Blommaert
	11:30-11:45	DNA methylation clock age acceleration as an indicator of environmental stress in king salmon	Ken Dodds
	11:45-12:00	Can machine learning models improve genomic prediction in Aquaculture? An application to a large dataset of Chinook salmon in New Zealand	Roy Costilla
	12:00-12:07	Flash Talk: Genomic information for increased prediction accuracy in New Zealand Wapiti deer	Jordan Clarke
	12:07-12:14	Flash Talk: From Quality Control to Biological Discovery: Lessons from Genetic Sex Validation	Chad Harland

12:15–13:15	**Lunch**		
13:15–14:15	**Session 2: Human** **Chair: Thomas Lopdell**		
	13:15-13:45	**Invited Speaker:** Resolving the Cellular and Molecular Architecture of the Human Olfactory Bulb Using Single-Nucleus and Spatial Transcriptomics	Alexandra Garton
	13:45-14:00	A Māori Pangenome 1.0 Grounded in a Tikanga and Mātauranga Māori framework	Phillip Wilcox
	14:00-14:15	Pleiotropic Identification for Aotearoa New Zealand Māori using Factor Analytic Linear Mixed Models	Alastair Lamont
14:15-15:00	**Session 3: Student flash talks – 5 min talk + 3 mins questions** **Chair: Steffi Fritsche**		
	14:15-14:23	Defining a longevity phenotype for genome-wide association studies in New Zealand dairy cows	Dameesha Abeygunawardana
	14:23-14:31	Automated coat pattern phenotyping to identify genetic modifiers of pigmentation in Holstein cattle	Melissa Wright
	14:31-14:39	A shared rumen core across sheep, beef and dairy cattle despite distinct community composition	Dermot Kennedy
	14:39-14:47	Can genomic selection improve heat tolerance and resilience of perennial ryegrass?	Arnel Pocsedio
	14:47-14:55	Males do it more - sex differences in recombination in the threatened hihi (stitchbird).	D'acy Searchfield

14:55–15:30	**Afternoon Tea**		
15:30–16:00	**Session 3 Continued: Student flash talks – 5 min talk + 3 mins questions** **Chair: Steffi Fritsche**		
	15:30-15:38	Cross-population gene prioritisation using conserved pathways and functional annotations	Florence Smith
	15:38-14:46	Investigating Population Structure Among NZ Kuku (Perna canaliculus) using Long Read Sequencing	Henry Collecutt
	15:46-15:54	Investigating the genomic basis of sex determination in the Green-lipped Mussel Perna canaliculus	Jamie Dalmer
	15:54-16:02	Dissection of a methane trait: exploring sustainability and productivity implications in sheep genetic evaluations	Shalini Abeykoon
	16:02-16:10	Genomic Signatures of Divergent Selection for Methane Yield in NZ Sheep	Zak Morrison
16:15-16:45	Bar Happy Hour		
16:45–17:00	Student Prizegiving – MapNet Organising Committee		
17:00-17:45	Mix and Mingle Bar still open		
18:00–late	Conference Dinner — Zenders, Hamilton		

Wednesday 4 November			
08:00–08:45	Arrive & coffee		
08:45–09:45	**Chair: Fenella Deans** **Keynote Speaker:** Ben Hayes - Potential approaches to create ultimate genotypes in crops and livestock		
9:45-10:15	**Session 4: Livestock** **Chair: Fenella Deans**		
	9:45-10:00	Breed of origin in genomic analysis of composite livestock populations	Andrew Hess
	10:00-10:15	Compound de novo KIT variants cause extreme coat depigmentation in a bovine sire family	Swati Jivanji
10:15–10:45	**Morning Tea**		
10:45–11:45	**Session 4 Continued: Livestock** **Chair: Fenella Deans**		
	10:45-11:00	Genomic evaluation of red deer milk traits in a New Zealand herd	Quinyi Luo
	11:00-11:15	Genetic Parameters for Fleece Shedding and Its Relationship with Growth Traits in New Zealand Sheep	Ruixin Liu
	11:15-11:30	Large-effect recessive variants impact sperm morphology and fertilisation traits in dairy cattle	Mathew Littlejohn
	11:30-11:45	How heritable is milk fatty acid composition in NZ dairy sheep?	Zongqi An
11:45-12:45	Steering Committee AGM / participant survey		
12:45–13:45	**Lunch**		

13:45–15:00	**Session 5: Plant Genomics** **Chair: Natalie Graham**		
	13:45-14:15	**Invited Speaker:** A story of genetics and trees	Heidi Dungey
	14:15-14:30	Pangenome construction and targeted sequencing panel for apple breeding	David Chagne
	14:30-14:45	Microhaplotypes improve genomic predictive ability in white clover	Hu Wang
	14:45-15:00	Multi-Omics Data Integration for the Discovery of Candidate Genes Involved in Dothistroma Needle Blight Resistance	Steffi Fritsche
15:00–15:45	**Afternoon Tea/Ice Cream**		
15:45–17:00	**Session 6: Genomics approaches and Computational Biology** **Chair: Chad Harland**		
	15:45-16:00	Beyond sequence: integrating structural data to reconstruct the evolution of bacteria movement	Caroline Puente-Lelievre
	16:00-16:15	Calling genomic variants at industrial scale	Frederic Bertels
	16:15-16:30	GWAS of Facial Eczema (FE) in NZ dairy cattle	Thomas Lopdell
	16:30-16:45	From SNP Chips to Sequencing: Paths to Low-Pass and Targeted Genotyping for Livestock and Horticulture	Chris Wicky
	16:45-17:00	What affects genomic offset predictions? Insights from a simulation model of the common myna invasion	Kamolphat Atsawawaranunt

17:00–17:45	Karakia whakamutunga, followed by board games & free evening
Thursday 5 November	
08:30–09:00	Meet-up
09:00–17:00	Innovation Farm tour (9:00-13:00) – packed lunch provided or Workshop hosted by REANNZ and Genomics Aotearoa – RNA-Seq DNA Analysis (9:00-15:30) – Morning Tea 10:30 and Lunch at 12:35
Afternoon	Departure

Abstracts

Talk 1: Keynote Speaker - Maren Wellenreuther
Breeding for an Uncertain Future

Modern breeding tools help us measure traits and select more precisely, but a changing environment raises a bigger question: are we selecting for the traits that will matter in the future? Yield remains essential, yet climate change, disease and environmental variability may change what is needed to sustain production.

We explore this question through our work on New Zealand snapper (tāmure), a warm-water species whose breeding programme is still closely connected to wild populations. Genomic and climatic data from 517 fish across 11 populations reveal temperature-associated genetic variation and predict the greatest genomic offsets in southern populations, including the region supplying the breeding programme.

We then tested how snapper from the breeding programme performed across four temperature regimes in a common-garden trial of around 3,200 F5 fish. Growth was highest in the constant optimal 21°C treatment, followed by the seasonally fluctuating warm (+2.5°C above ambient), ambient (the five-year average for Nelson) and cold treatments (−2.5°C below ambient). Warmer temperatures increased growth rates, and pedigree analysis indicated high additive genetic variation for final weight. Families also varied in growth-rank stability and their response to a short simulated heatwave, but these measures were not associated in this trial.

Together, these findings show how wild-population genomics and experimental trials can help assess future environmental challenges. They also highlight the work ahead: measuring resilience and robustness reliably, and determining how these traits can inform breeding selection alongside yield. This will help us develop breeding objectives that maintain production as conditions change.

Speaker Biography

Professor Maren Wellenreuther is a Professor at the University of Auckland and Science Group Leader at the Bioeconomy Science Institute. Her research focuses on evolutionary biology, fisheries genomics, and aquaculture, with an emphasis on improving the sustainability and resilience of aquatic ecosystems. She has published over 130 peer-reviewed papers, leads a research group of more than 30 scientists and students, and serves as Editor-in-Chief of *Evolutionary Applications*. Her achievements have been recognised through several international honours, including a Humboldt Fellowship and the Royal Society Te Apārangi Early Career Research Excellence Award.

Talk 2: Genomic erosion in the threatened hihi / stitchbird

Anna Santure[1], Hui Zhen Tan*[1], Patricia Brekke[2]

[1]School of Biological Sciences, Waipapa Taumata Rau - University of Auckland, 3 Symonds Street, Auckland

[2]Institute of Zoology, Zoological Society of London, Regent's Park, London

*Lead Author

_ Is the presenting author

Small, threatened populations are vulnerable to genomic erosion, where reduced genetic diversity and increased inbreeding reduce adaptive potential and elevate extinction risk. Population genomic analyses can quantify these processes and inform conservation management. Hihi/stitchbird (*Notiomystis cincta*) is a forest bird endemic to Aotearoa New Zealand. Once widespread across the North Island, the species was reduced to a single remnant population on Te Hauturu-o-Toi following habitat loss and the introduction of mammalian predators. Conservation translocations have since established seven additional populations. Our previous work has revealed low genetic diversity, high inbreeding, and inbreeding depression in the reintroduced Tiritiri Matangi population, but genomic variation across other populations has not been assessed. Here, we combined low-coverage whole-genome sequencing with imputation from a high-coverage reference panel to generate genome-wide SNP data for 467 individuals from Te Hauturu-o-Toi, Tiritiri Matangi, and the severely bottlenecked Zealandia population. We estimated an effective population size of approximately 2,000 individuals on Te Hauturu-o-Toi, similar to its census size. However, both reintroduced populations showed evidence of genomic erosion, with Zealandia exhibiting substantially reduced genetic diversity and elevated inbreeding. First-generation offspring from crosses between Te Hauturu-o-Toi and Tiritiri Matangi individuals had the highest genetic diversity and lowest inbreeding levels, suggesting that sourcing founders from multiple populations could mitigate genomic erosion in future translocations. Our results demonstrate substantial variation in genomic erosion among hihi populations and highlight the importance of population-wide genomic monitoring to guide conservation management and prioritise interventions.

Talk 3: Genomic origins and population structures of the Huntaway and Heading Dog breeds in New Zealand

Milly Henry[1*], Melissa Stephen[1], Florence Smith[1], Gareth Enticott[2], Hayley Hunt[3], Nick Sneddon[1], Keren Dittmer[3], Dorian Garrick[4], Garry Mainland[1], Thomas Lopdell[1] & Mathew Littlejohn[1]

[1] Massey University, School of Agriculture & Environment
[2] Cardiff University
[3] Massey University, School of Veterinary Sciences
[4] The Helical Company
*Lead Author
_ Is the presenting author

Background

Working dog breeds were developed with a focus on traits required for specific tasks such as herding or guarding, with little regard to aesthetic traits. Two breeds of working farm dogs were developed in New Zealand (NZ) and remain largely specific to the country: the Huntaway and the Heading dog. Despite their importance to agriculture, limited scientific research has been conducted to uncover the origins, development, and genetic composition of these breeds. Here, we provide the first large-scale genomic investigation of the Huntaway and Heading Dog populations.

Results

We analyse the population structure of 920 Huntaways and 842 Heading Dogs and compare them with 1,256 international canids, representing 127 dog breeds and three wild canid species. Using a panel of 151,688 variants common to all dogs, we establish the genomic composition and ancestry of these populations. The results demonstrate that both breeds can be traced back to ancestral United Kingdom (UK) rural working dog populations. Analyses also show that, while the Heading Dog is closely related to the Border Collie, the population is genetically distinct – presumably due to specific selection pressures unique to NZ and the functionally specialised roles they play on farms.

Conclusions

This study provides genomic evidence that the Huntaway and Heading Dog share ancestry with UK rural working dog populations. Both breeds occupy a position within the UK rural subclade on the domestic canine phylogeny, a finding that is consistent with historical patterns of European settlement and dog importation during the agricultural expansion of NZ in the 19th century.

Talk 4: Structural variant size, type and genomic context shape continuous growth variation in snapper (*Chrysophrys auratus*)

Julie Blommaert[1], Samuel Vander Velpen[1], Linley Jesson[2], Maren Wellenreuther[1,3]

[1] New Zealand Institute for Bioeconomy Science, Nelson, New Zealand

[2] New Zealand Institute for Bioeconomy Science, Lincoln, New Zealand

[3] School of Biological Sciences, The University of Auckland, Auckland, New Zealand

_ Is the presenting author

Most genomic breeding evaluations focus on single nucleotide polymorphisms, but structural variants (SVs) have also been shown to impact phenotypes, often drastically. Studies are beginning to focus on the contribution of SVs to continuous, polygenic traits, an example being growth. Growth is one of the most important traits in selective-breeding programmes, yet the contribution of SVs to growth variation remains poorly resolved. Australasian snapper (*Chrysophrys auratus*) is an emerging aquaculture species in New Zealand with an established selective-breeding programme and existing genomic resources, making it possible to test whether SVs contribute to quantitative growth traits. We examined SVs associated with body weight 100 using whole genome sequenced F_4 snapper (mean coverage 9.3x). We identified over 80,0000 SVs using three complimentary calling programmes. We prioritised candidate SVs using a presence-absence based phenotype-genotype association pipeline as well as allele dosage based penalised regression. Through the phenotype-genotype association pipeline, we identified 239 variants, and using penalised regression, we identified 925 variants associated with weight and identified that association was not linked with SV size. Direct overlap between candidate SVs from both analyses was limited, but over 50% of SVs intersected with genes (association candidates: 57.3%; regression candidates: 58.4%). Both analyses also identified the same high SV-density chromosomes (chromosomes 11 and 22). This adds to a growing body of work showing that structural variants can shape complex polygenic traits and suggests that similar approaches could help uncover hidden genomic variation relevant to continuous traits important for breeding, adaptation and conservation in other species.

Talk 5: DNA methylation clock age acceleration as an indicator of environmental stress in king salmon

Dodds, K.G.[1], Clarke, S.M.[1], Brauning, R.[1], French, M.C[1]., van Stijn, T. C.[1], Waddington, Z[2], Symonds, J E[3]

[1]Bioeconomy Science Institute, Invermay Agricultural Centre, Mosgiel, NZ
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[3]Cawthron Institute, Nelson, New Zealand
_ Is the presenting author

DNA methylation profiles can be used to predict the biological age of a sample via a "methylation clock". This prediction uses a training set of individuals from a "normal' environment. Deviations of the methylation age from the chronological age (known as age acceleration) may reflect stress due to a harsh environment. A chinook (king) salmon (*Oncorhynchus tshawytscha)* clock was developed from fin clip samples collected from either a routine breeding programme sample, or when entering a research trial. Subsequently, fin clips were collected from 92 fish at up to four timepoints during a heat-stress trial. Both sets of fish were assayed for methylation levels using the Reduced Representation Enzymatic Methyl-Sequencing (RREM-seq) method. To develop the methylation clock, only CpG sites (n≈ 200,000) with at least five reads in all training samples were retained. A clock was developed using elastic net with response variable ln(age + 60 days) which had a leave one out cross-validation correlation of 0.989 and selected 103 sites for prediction. Age acceleration in the heat-stress trial samples was analysed with timepoint as a factor. Age acceleration significantly increased from tagging to pre-heat-stress sample and to mortality or post-heat-stress sample. There was no significant additional change from then to the post-recovery sample. These results are preliminary evidence that a methylation clock can be used as an indicator of environmental stress in chinook salmon.

Talk 6: Can machine learning models improve genomic prediction in Aquaculture? An application to a large dataset of Chinook salmon in New Zealand

Roy Costilla[1,2], Megan Scholtens[1], Shannon M. Clarke[3], Ken G. Dodds[3], Zac Waddington[4], Ivy Liu[2], and Jane E. Symonds[1].

[1]Cawthron Institute, Nelson, New Zealand
[2]School of Mathematics and Statistics, Victoria University of Wellington, New Zealand
[3]Invermay Agricultural Centre, Bioeconomy Research Institute, Mosgiel, NZ
[4]The New Zealand King Salmon Company, Nelson, New Zealand.
_ Is the presenting author

Selective breeding incorporating genomic information is used for several aquaculture species in New Zealand, including Chinook salmon (*Oncorhynchus tschawytscha*). In recent years, machine learning (ML) models have risen as black-box prediction tools which promised to capture non-linear genomic relationships and improve prediction. However, well-powered comparisons of conventional breeding and ML models in aquaculture are lacking due to the large sample sizes required to accurately estimate genetic effects for complex traits. Such comparisons are needed because the genomic best linear unbiased prediction (GBLUP) model, the foundational model for genomic prediction, assumes an infinitesimal genetic architecture where all genetic markers contribute to the trait variance. So far GBLUP still matches or outperforms other prediction methods. Here, we compare the GBLUP model with several ML models as well as other Bayesian models often used in selective breeding. We used genotyping by sequencing (GBS) data from 4,924 samples for three growth traits in Chinook salmon: weight, fork length and growth rate. We evaluated the performance among models using ten-fold cross-validation, defining accuracy as the correlation between real and predicted phenotypes adjusted by the trait heritability. We identified 9,118 single nucleotide polymorphisms (SNPs) with a mean depth=3.04 and a call rate=0.90. Estimated heritabilities were high (weight and fork length) to moderate (growth rate). Prediction accuracies were very similar across the models and traits. We therefore found no significant improvement in prediction accuracy when using the ML models. On the other hand, accuracies weren't much lower either for the ML models which could be advantageous as these models do not necessitate any prior assumptions about the underlying genetic trait architecture. In conclusion, by using a large dataset of real-world salmon data, we highlight the potential of ML models for prediction of genetic values and provide insights into their strengths and limitations in aquaculture.

Keywords: Non-additive genetic models, neural networks, decision trees, BayesC, BayesR.

Talk 7: Genomic information for increased prediction accuracy in New Zealand Wapiti deer.

J.L. Clarke[1*], K.G. Dodds[1]

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Deer are an underutilised agricultural species, capable of occupying a wide range of niches. Deer are primarily farmed for venison and velvet, as well as trophy hunting and milk. However, they are not as established as other agricultural species, particularly regarding integration of genomics in breeding value estimation. The use of genomic relationship matrices in deer breeding selection using low-cost genotype by sequencing (GBS) methods has been established although has not been incorporated into national evaluations. Current animal selection in New Zealand deer largely relies on pedigree information alone, despite genotyping of key individuals becoming more prevalent. A genomics-based evaluation needs to retain the non-genotyped individuals in addition to utilising genomic information. Furthermore, investigation of genomic prediction of deer has to date been limited to a few key traits. This study analyses the performance of genomic and pedigree-based models for prediction of key economic traits in New Zealand wapiti using Mix99 software (as is currently used for national evaluations), and the accuracy and efficiency of both methods. The aim of the project is to implement sustainable genomic selection practices in the New Zealand deer industry, particularly for wapiti subspecies.

Talk 8: From Quality Control to Biological Discovery: Lessons from Genetic Sex Validation

Katie Carnie[1], Fenella Deans[1], Chad Harland[1]

[1]Livestock Improvement Corporation (LIC), Research and Development, Hamilton, New Zealand
_ Is the presenting author

As genomic information becomes increasingly central to breeding and research programmes, robust quality assurance (QA) processes are essential to ensure that only high-quality data are used for downstream analyses and selection decisions. Effective QA extends beyond standard sample and laboratory metrics, such as DNA quality, instrument performance, genotype quality, and call rate, to include animal-level validation measures including global duplicate detection, genotype discordance, breed or line assignment, parentage identification, and sex determination.

In this presentation, we highlight how genetic sex determination can provide a powerful additional layer of QA. Through a recent real-world example from our routine genomic QA pipeline, we demonstrate how sex prediction, alongside other genomic validation checks, can identify issues and uncommon biological phenomena. This case study illustrates the value of comprehensive genomic QA pipelines in maintaining data integrity while also uncovering rare and informative biological edge cases.

Talk 9: Invited Speaker – Alexandra Garton
Resolving the Cellular and Molecular Architecture of the Human Olfactory Bulb Using Single-Nucleus and Spatial Transcriptomics

Olfactory dysfunction occurs in more than 90% of Parkinson's disease (PD) patients, often preceding motor symptom onset by 4–10 years. Additionally, pathological changes are evident in the olfactory bulb (OB) early in PD. However, the molecular and cellular organisation of the human OB remains largely uncharacterised, limiting insights into these early pathogenic events. A comprehensive molecular characterisation of the human OB is therefore needed to establish a reference for investigating these disease-associated changes. Here, we characterise the cellular and spatial organisation of the human OB using single-nucleus RNA sequencing (snRNA-seq) and spatial transcriptomics.

SnRNA-seq of postmortem OB tissue from six donors (two with PD and four neurologically normal controls) identified 21 transcriptionally distinct cell populations across 86,461 nuclei, comprising 12 neuronal and nine non-neuronal populations. Spatial transcriptomics of fresh-frozen OB sections using 10x Genomics Visium Spatial Gene Expression (four sections) and formalin-fixed, paraffin-embedded (FFPE) OB sections using Visium HD (seven sections) resolved the spatial and layer-specific distributions of these populations. Unsupervised clustering of Visium data revealed spatial patterns corresponding to the major anatomical layers of the OB. Deconvolution of Visium HD data using the snRNA-seq dataset as a reference enabled spatial localisation of the 21 identified cell populations, including a novel CDH19+/ LEF1+ population that was found to correlate with olfactory glomeruli. Integration of orthogonal transcriptomic technologies supports the presence of two transcriptionally and spatially distinct glutamatergic populations and four transcriptionally distinct granule cell populations. Immunohistochemistry has provided initial validation of candidate markers for distinct glutamatergic populations, including ENC1, NEUROD2 and EOMES. Preliminary comparisons between PD and control tissue have also identified a candidate granule cell population overrepresented in PD cases that warrants further investigation.

Together, these findings provide a molecular and spatial foundation for defining the cellular organisation of the human OB and support future investigations of cell-type-specific changes and potential cellular vulnerability in PD.

Speaker Biography

Alexandra Garton is a postdoctoral fellow at the University of Auckland. She completed her PhD in 2025, where she used multi-omics approaches, including single-nucleus RNA sequencing and spatial transcriptomics, to study the human olfactory bulb and its role in Parkinson's disease. Her research now focuses on spatial transcriptomics and proteomics technologies to better understand disease-associated cell states and molecular changes in the brain.

Talk 10: A Māori Pangenome 1.0 Grounded in a Tikanga and Mātauranga Māori framework

Ann M. Mc Cartney[1,2], Benjamin J. Halliday[3,4], David M. Markie[5,4], Julian K. Lucas[2], Brandy M. McNulty[2], Elizabeth Goodin[3,4], Ivo Violich[2], Joshua Gardner[2], Tyler McInnes[4], Dinindu Senanayake[6], Jian Sheng Boey[4], Matt Bixley[6], Brandon Pickett[7], Linelle Ann L. Abueg[8], Todd Hillaker[2], Celine Neudorf[2], Helen Wihongi[9,10,30], Anna Rolleston[11,12,30], Maui Hudson[13,14,30], Irene Kereama-Royal[15,16,30], Andrew Sporle[17,18,30], Jason Alexander[19,20,30], Nadine Hapuku[21,22,30], Ben Te Aika[23,24,30], Jennie Harré -Hindmarsh[30], Tuta Haereroa[25,26,30], Huti Puketapu-Watson[25,27,30], Stephen P. Robertson[3,4], and Karen H. Miga[2], Phillip L. Wilcox[28,29]

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Recently a 1.0 version of a Māori pangenome was created by integrating six newly constructed Māori telomere-to-telomere (T2T) reference genomes into a pangenome alongside assemblies produced by the Human Pangenome Reference Consortium (HPRC). This project was enabled via specific tikanga Māori 'tauutuutu' ethical framework that underpinned the creation of this pangenomes via an international collaboration between the He Kākano project team and Prof Karen Miga's group at the University of California Santa Cruz. The first step in the pangenome was constructing six individual telomere-to-telomere assemblies from a combination of Nanopore-generated long read data along with PacBio HiFi and HiC data. We integrated these six Māori reference genomes with the 231 HPRC2 genomes using CHM13 and GRCh38 as reference sequences, resulting in a graph comprising 476 haplotypes. As an initial evaluation of the contribution of the Māori assemblies, we assessed representation of coding variants that are globally rare (gnomAD v4.1 AF ≤ 0.0001) but common among Māori (He Kākano Variome v1.0.1 AF ≥ 0.01) as a proxy for Māori-specific variation. Of the 3,056 such coding SNPs identified, 638 (20.9%) were present in He Kākano + HPRC2, compared with only 96 (3.1%) in HPRC2 alone. Further evaluations will be described in the presentation. The Māori pangenome V1.0 will be made available for research and clinical purposes under specific conditions as determined by the He Kākano Māori leadership roopu. Through this collaboration, we have showcased how it is possible to ground genomics projects in Indigenous ethical principles and protocols while also producing resources that are comparable in quality and interoperable with leading genomics projects across the globe.

Talk 11: Pleiotropic Identification for Aotearoa New Zealand Māori using Factor Analytic Linear Mixed Models

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Traditionally, genome-wide association studies (GWAS) have been pivotal in linking genetic variants to disease risk, uncovering relevant molecular pathways, and providing targets for repurposing drugs. Identifying pleiotropic variants – loci which influence multiple traits – may reveal insight into fundamental biological processes and inform treatments that simultaneously improve multiple health conditions.

New Zealand Māori have unique genetic architecture that GWAS in other populations do not capture. Standard GWAS requires large, expensive datasets of $\geq 10^5$ individuals. This means it is not an ideal tool for pleiotropic identification in datasets focused on under-represented populations such as the Genetics of Gout, Diabetes and Kidney Disease (GoGDK) tāonga – which contains the genetics over 2000 individuals of Polynesian descent including Māori, measured for a wide range of metabolic conditions including gout, type 2 diabetes and renal function.

We have instead used Factor Analytic Linear Mixed Models (FALMM), an extension of single-trait Linear Mixed Models that treats the covariance between traits as a combination of trait-specific variances plus some number of underlying causal factors that influence multiple traits - a pleiotropic component. These models have the advantages that they a) stably converge even with small datasets b) can be interpreted for common effects and c) can be solved with efficient algorithms. Genes identified thus far are congruent with previously discovered genetic variants. This method holds promise for future precision medicine applications for small populations.

Talk 12: Student Flash Talk - Defining a longevity phenotype for genome-wide association studies in New Zealand dairy cows

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Longevity is an important trait for the economic and environmental sustainability of pasture-based dairying but is difficult to define for GWAS because many cows are alive at data extraction or leave the herd without ending their productive life, resulting in censored records. As phenotype choice determines which cows and information enter GWAS, this study aimed to develop a GWAS phenotype making best use of censored records from 6.5 million New Zealand cows born 2005 – 2015. Removal by culling, death or sale for a specified reason was the event, cows alive at data extraction or sold for export or as surplus were censored (1 - 14% across birth years). Herd life (HL) and productive life (PL) were analysed with linear mixed models (LMM), using full and restricted data, and Cox proportional hazards models, including breed, herd at first calving (random effect in LMM, fixed effect in Cox) and age at first calving relative to herd × birth-year contemporaries, birth year was fixed effect in the LMM and a stratum in the Cox model. Follow-up time was equalised at the youngest birth year cohort's maximum possible lifetime (HL 3,466 d, PL 2,369 d) cows exceeding these thresholds were excluded from the restricted LMM and administratively censored in the Cox model. Crossbred cows survived longer than purebred cows in all analyses. Friesian × Jersey (F×J) crossbreds had the longest HL and PL in the full-data LMM (138 d longer than purebred Friesians for both traits) and the lowest hazard of removal in the Cox model (HR = 0.862 for HL and 0.857 for PL). In the restricted LMM, however, their advantage fell to 76 d (HL) and 60 d (PL), F×J and Jersey crossbreds no longer differed, and purebred Jerseys ranked above purebred Friesians, unlike in the full-data LMM and Cox model. The handling of censored records thus altered effect sizes and rankings. Because the full-data LMM treats censored records as complete, two other candidate GWAS phenotypes were derived, the restricted LMM residual (excludes long-lived and still-alive cows) and the Martingale residual from the Cox model, defined as a cow's observed event status minus her estimated cumulative hazard. The Martingale residual retains all 6.5 million cows (LMM residual - 4.76 million). Both will be compared on distribution, correlation, heritability and detected loci to identify the best phenotype for GWAS of additive and large-effect, non-additive variants.

Talk 13: Student Flash Talk - Automated coat pattern phenotyping to identify genetic modifiers of pigmentation in Holstein cattle

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Heat stress poses a growing threat to the livestock industry, negatively impacting productivity and welfare. Coat pigmentation influences heat tolerance in dairy cattle, particularly in Holsteins, whose black-and-white pattern is of genetic and welfare interest. Black regions of the coat absorb more solar radiation, increasing susceptibility to heat stress, whereas white regions lack protective pigment and are more susceptible to ultraviolet radiation-induced skin damage. Identifying and selecting genetic variants underlying heat tolerance-related pigmentation patterns may aid herd adaptation to a warming climate. The extent of skin pigmentation is determined largely during embryonic development by pigment-producing cells called melanoblasts, which develop into melanocytes. Despite substantial variation in Holstein pigmentation patterns, only three genes, all involved in melanoblast development (*MITF*, *KIT* and *PAX3*), have thus far been identified as influential, suggesting that further modifiers remain undiscovered. Routine genotyping in the dairy industry presents an opportunity to map these modifiers, but this requires robust phenotyping of pigmentation patterns at scale. Here, we aim to develop a semi-automated computer vision pipeline for large mammal pattern analysis. This will enable image analysis to define pigmentation phenotypes, improving the power of genome-wide association studies (GWAS) to identify novel associated variants. In parallel, body temperature data from ear tags and thermal imaging will be used to identify pigmentation patterns associated with reduced thermal load and better heat tolerance. Overall, this study presents a framework for characterising novel coat-pattern traits, identifying their genetic basis, and informing breeding strategies to improve resilience and welfare under a changing climate.

Talk 14: Student Flash Talk - A shared rumen core across sheep, beef and dairy cattle despite distinct community composition

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The rumen microbiome drives methane production and feed digestion in ruminants, yet how consistently microbial taxa occur across production populations is unclear. We characterised rumen community structure across sheep, beef cattle and dairy cattle, and quantified how study and individual animal contribute to variation in shared taxa.

We combined 1,535 rumen samples from 1,316 animals across 10 study populations (1,115 sheep, 121 beef and 299 dairy samples). Communities were profiled by 16S rRNA gene sequencing (DADA2, SILVA 138) and aggregated to genus level. Shared-core genera were defined as those detected in at least 80% of samples within every host group, and their abundance variation was partitioned among study, animal and residual effects.

Overall composition differed among host groups (PERMANOVA P = 0.001), with host group explaining 45.9% of variation among study-level profiles, although richness and diversity did not differ. Nevertheless, 77 genera formed a shared core, including Prevotella, Ruminococcus, Fibrobacter and Methanobrevibacter. Study explained a median 14.1% of remaining variation in these taxa, compared with 1.9% for animal, though animal variation exceeded 50% in eight genera. Only three core genera differed significantly among host groups after correction.

Contrasting ruminant populations therefore share a substantial set of highly prevalent taxa, but widespread occurrence does not guarantee consistent abundance. Variance partitioning identifies which taxa are sensitive to study conditions and which reflect between-animal differences. The next step is testing whether these conserved features retain consistent associations with methane and other traits across independent populations, as required for their use in breeding.

Talk 15: Student Flash Talk - Can genomic selection improve heat tolerance and resilience of perennial ryegrass?

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Declining productivity and persistence in perennial ryegrass (PRG)-based pastures have been linked with rising atmospheric temperatures. Developing heat-tolerant cultivars with strong recovery capacity is therefore important to mitigate the impacts of a warming climate. This study evaluated the potential of genomic selection to support breeding for heat tolerance and post-stress recovery in PRG. We screened 226 diverse NZ-adapted PRG genotypes under heat stress (38°C/30°C, day/night) and control (23°C/15°C) conditions. Dry matter yield (DMY), plant height (PH), and visual scores of heat damage and regrowth were assessed across three phases: two weeks of heat stress (P1), five weeks of cumulative stress (P2), and post-stress recovery (P3). DMY data were used to calculate stress tolerance (STI) and relative stress tolerance (RSI) indices, which rank genotypes for relative stress susceptibility and yield stability. Using a proven genotyping-by-sequencing dataset, we built a SNP-based genomic relationship matrix for 209 genotypes and integrated it with phenotype data to fit genomic best linear unbiased prediction models. Preliminary cross-validated predictive abilities (PA) were low to moderate for DMY, with the highest PA observed in P1 under both control (0.29) and heat-stress (0.27) conditions. PH generally showed higher PA (0.16–0.48), whereas visual scores showed lower PA (0.15–0.24). Among the stress indices, RSI_P3 had the highest PA (0.37), followed by RSI_P2 (0.34) and STI_P1 (0.28). These interim findings reveal genotypic variation for heat-stress response in NZ-adapted PRG and demonstrate the potential of genomic selection to accelerate the development of resilient cultivars for increasingly warm and variable production environments.

Talk 16: Student Flash Talk - Males do it more - sex differences in recombination in the threatened hihi (stitchbird)

D'Acy Searchfield[1], Phoebe Scherer[1], Hui Zhen Tan[1], Kate D Lee[1], Patricia Brekke[2], Anna W Santure[1]

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Recombination is a near-universal process in sexually reproducing organisms that ensures proper chromosome segregation during meiosis and helps maintain genetic diversity. The recombination landscape, characterised by the rate and distribution of crossover events, varies both among and within species. One common form of within-species variation is heterochiasmy, whereby recombination patterns differ between sexes. Although heterochiasmy has been identified across a range of taxa, the mechanisms driving this variation remain poorly understood.

Hihi (stitchbird; *Notiomystis cincta*), an endangered Aotearoa, New Zealand passerine, exhibits male-biased heterochiasmy. We further investigated hihi heterochiasmy by quantifying individual variation in recombination traits and characterising sex-specific recombination patterns. Linkage map constructions for 30 chromosomes were created in CRIMAP from 8,226 SNPs genotyped across 1,009 individuals from 24 families. Using 612 informative meioses, autosomal crossover count (ACC) and intrachromosomal shuffling (rIntra) were estimated for 105 females and 92 males based on the average number of recombination events inherited by their offspring.

Both ACC and rIntra show significant difference between sexes, with males having a 1.39 times higher ACC, and 1.21 times higher rIntra compared to females. The heritability of ACC was low for both sexes, though higher in males, and near-zero for rIntra. Genome-wide association found no loci associated with ACC. These results provide further characterisation of heterochiasmy in hihi and demonstrate differences in sexes across a number of recombination traits. This work contributes to furthering understanding of the mechanisms and evolutionary basis of sex-specific recombination variation.

Talk 17: Student Flash Talk - Cross-population gene prioritisation using conserved pathways and functional annotations

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Genome-wide association studies (GWAS) and functional analyses have generated extensive genetic databases for humans, especially for populations with European ancestry. These datasets could be leveraged to improve the identification of causal genes in less well studied populations and species. For example, height GWAS may improve the identification of loci for stature in cattle. However, there is no well-established framework for quantifying and exploiting shared pathways across populations.

We aim to develop a novel cross-population pipeline for identifying causal genes that leverages conserved biological pathways. We extend an existing gene prioritisation model to incorporate multi-population association statistics and annotations in a Bayesian statistical framework. Initial testing in human height GWAS across ancestries suggests our framework has a higher prediction accuracy than other methods in underpowered studies. Our framework was also able to correctly prioritise relevant height genes with greater precision than existing approaches in small cohorts. Further, we find evidence that effects are shared across ancestries, supporting the potential utility of conserved features across populations. However, the cross-ancestry model did not outperform the single-ancestry model, potentially due to saturation of genetic signals in pathways for height. We will test this hypothesis using smaller cohorts and alternative traits.

Next, we will optimise the model for cross-species analyses, then apply the method to New Zealand dairy cattle by incorporating human and cattle studies with the aim to identify shared signals underlying analogous traits. This work could provide evidence for conserved biology and establish whether this can be exploited to prioritise candidate causal genes.

Talk 18: Student Flash Talk - Investigating Population Structure Among NZ Kuku (Perna canaliculus) using Long Read Sequencing

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Kuku, or green-lipped mussels, have a high degree of genomic hemizygosity, a state in which the copy number of genes differs between populations and individuals [1]. Hemizygosity has been previously shown in other species, including bivalves, to exhibit significant differences between geographically separated populations, and annotation of genes under presence/absence variation in these regions is often returns functions related to stress response, including immunity, calcium regulation and temperature [1].

As an economically [2], ecologically [3][4] and culturally significant taonga species, Kuku adaptation in response to a changing climate is of keen interest. Green-lipped mussels are under threat, as rising temperatures, ocean acidification [5], and sedimentation, as well as increasing frequency of El Niño [6][7], all pose dire risks for the endurance of NZ kuku.

30 individuals were sampled from four different locations throughout the South Island; Kaka Point, St Kilda, Ōnuku and Karaka Point. High molecular weight DNA was extracted from the abductor muscle tissue using spin-column based DNA extraction and prepared in barcoded libraries for sequencing using the ONT PromethION flow cell. The target N50 for sequenced genomes was ~10-20kb, with a minimum of 5x coverage.

Preliminary sequencing and assembly has produced four individual mussel genomes, from individuals sampled in Dunedin, Karaka Point and Ōnuku. Future sequencing of the remaining individuals, combined with extant genomic data from North Island lineages [8] will be used to construct a Kuku pangenome to quantify structural variation across NZ Kuku and annotate genes with presence/absence variation in each population to resolve the potential relationship between hemizygosity and climate adaptability.

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Talk 19: Student Flash Talk - Investigating the genomic basis of sex determination in the Green-lipped Mussel Perna canaliculus

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The Green-lipped Mussel, traded under the Greenshell™ name, is a bivalve mollusc which has great ecological, cultural, and economic significance in New Zealand. Greenshell™ mussels are a key aquaculture species in the growing blue economy, with their cultivation contributing over $400 million in revenue annually[1]. As with many species in the class Bivalvia, the process by which biological sex is determined in this species remains undescribed. Previous studies have demonstrated the efficacy of investigating sex determination using the growing suite of genomic tools available to researchers [2,3].

To this end, the genomes of 12 male and 12 female individuals, from samples taken in partnership within an iwi-lead breeding programme, were re-sequenced using Illumina short-read sequencing. These sexually stratified genomes were screened for regions of consistent sex-based differentiation indicative of a role in sex determination. To identify these regions, read mapping rates to the male reference genome, SNP-based FST and sex-specific heterozygosity were used as measures of sexual differentiation.

Based on these analyses a putative female-specific 150kb region was identified which showed high FST and exclusive heterozygosity. Further verification is required to determine whether this region is linked to sex determination and to assess if it shows promise as a sex-specific genomic marker[2]. These findings contribute towards the growing selective breeding efforts being undertaken in the industry[4], potentially facilitating earlier and increasingly accurate mussel sexing. Our data will move forward to implementation frameworks in partnership with kaitiaki, who will hold this knowledge in perpetuity for use as they see fit.

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Talk 20: Student Flash Talk - Dissection of a methane trait: exploring sustainability and productivity implications in sheep genetic evaluations

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Internationally, methodologies employed for ranking animals on their greenhouse gas output have been a contentious topic. Any method decided upon needs to take into consideration other factors such as whether feed supply is fixed or variable, preserving productivity traits, and being realistic on the levels of feed intake and gaseous phenotypic data sources.

This study develops a deterministic selection-index framework in which genetic variation in daily methane is partitioned into three biological components: methane associated with feed intake required for productive output, methane associated with residual feed intake, and methane associated with methane yield. Daily feed intake is similarly decomposed into production-associated and residual components. Genetic variances and covariances are parameterised using path-coefficient methodology and literature-derived genetic parameters, allowing alternative assumptions regarding the proportion of methane variation attributable to each evaluated pathway.

Results will evaluate whether a classical definition of linear selection indexes leads to unexpected and unwanted outcomes when there is failure to conceptionally separate a single but multifaceted trait into components that differ in their individual value. This has further relevance when modelling the potential value of alternative methane measurements to understand how direct selection towards inherent methane efficiency can reduce overall methane production whilst preserving the productivity of animals.

Talk 21: Student Flash Talk - Genomic Signatures of Divergent Selection for Methane Yield in NZ Sheep

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New Zealand is heavily dependent on animal agriculture, with combined lamb, mutton, and wool exports forecasted at NZ$4.26 billion in 2026. Enteric methane from grazing livestock is, however, a major contributor to national emissions, and a potential threat to the economic sustainability of the primary sector.

Breeding for reduced methane, in contrast to other proposed solutions, is permanent and cumulative, and has begun to be incorporated into national genetic evaluations. It remains poorly understood which genes are responsible for variation in methane, and how long-term selection might affect other production traits.

In this work, we characterised signatures of selection in flocks of sheep divergently selected for high and low methane yield over fourteen years. Differentiation between lines was estimated using fixation indices, with simulation testing whether values exceed expectations under a neutral evolutionary model. We also used methods based on extended haplotype homozygosity and nucleotide diversity (iHH_{12}, XP-EHH, θ_π), identifying candidate regions under recent positive selection both within and between lines.

We identified evidence of selection at several loci overlapping quantitative trait loci (QTL) previously associated with lipid metabolism and immune function, with further annotation and gene ontology analysis linking these regions to candidate genes and biological pathways.

Talk 22: Keynote Speaker – Ben Hayes
Potential approaches to create ultimate genotypes in crops and livestock

Speaker Biography

Professor Hayes has extensive research experience in genetic improvement of livestock, crop, pasture and aquaculture species, with a focus on integration of genomic information into breeding programs, including leading many large-scale projects which have successfully implemented genomic technologies in livestock and cropping industries.

Author of more than 300 journal papers, including in Nature Genetics, Nature Reviews Genetics, and Science, contributing to statistical methodology for genomic, microbiome and metagenomic profile predictions, quantitative genetics including knowledge of genetic mechanisms underlying complex traits, and development of bioinformatics pipelines for sequence analysis. Thomson Reuters highly cited researcher in 2015, 2016, 2017 and 2018.

Talk 23: Breed of origin in genomic analysis of composite livestock populations

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In composite and crossbred livestock populations, genomic regions can differ in their ancestral breed of origin within and among animals, creating challenges for interpreting genomic associations. The same marker allele can occur on different ancestral breed backgrounds, and trait-associated regions may reflect breed composition, local ancestry, or true within-breed genetic effects.

Wool shedding in New Zealand sheep provides a useful example. Wiltshire genetics have been introduced into non-shedding populations to improve shedding performance, creating populations in which phenotype, breed ancestry, family structure and genomic variation are closely related. Standard genomic analyses can identify regions associated with a trait, but they do not necessarily reveal the ancestral origin of those regions or whether an association is expected to transfer across composite populations.

This presentation will use shedding sheep as a case study to outline the analytical questions that arise when traits are introduced through breed introgression. We will consider the roles of global ancestry, local ancestry and within-breed genetic variation, and how these can be separated conceptually in genomic analyses. Local ancestry inference provides one route toward identifying the breed of origin of genomic regions and testing whether genetic effects are shared across ancestral backgrounds.

The broader aim is to improve interpretation of genomic results in admixed livestock populations and to distinguish association with breed ancestry from association with transferable genetic effects.

Talk 24: Compound de novo KIT variants cause extreme coat depigmentation in a bovine sire family

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This study presents an investigation of extreme depigmentation phenotypes observed in all progeny from a single, seemingly unaffected Hereford sire. Short and long-read sequencing identified two distinct, overlapping heterozygous structural variants disrupting coding regions of the *KIT* gene, a key regulator of melanoblast development. These variants segregated on different paternal haplotypes, suggesting germline mosaicism and a *de novo* origin for both mutations. These observations highlight erroneous repair of a single DNA lesion as the likely mechanism of mutagenesis, resulting in an unusual inheritance pattern whereby *de novo*, haploinsufficient alleles are transmitted to all progeny.

Talk 25: Genomic evaluation of red deer milk traits in a New Zealand herd

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Red deer (*Cervus elaphus* L.) are emerging as a commercial dairy species in New Zealand, yet no genomic evaluation of milk traits exists for any deer species. This gap leaves an expanding industry without genomic tools for selection and no example for a novel ruminant.

This study presents the first genomic breeding values (GEBVs) for six milk production and udder health traits (milk, fat, protein and lactose yields, lactation persistency and somatic cell score), from 319 test-day records across two lactations from 219 hinds and 834 genotyped animals. Genotyping-by-sequencing yielded 46,279 autosomal SNPs, used to build a read-depth-aware genomic relationship matrix suited to the low and uneven coverage of GBS. Deer-specific variance components were unavailable, so heritabilities and repeatabilities were adopted from published dairy sheep evaluations, and variance components fixed in a single-trait repeatability GBLUP model.

Mean GEBV accuracies were 0.45–0.63 (lowest for lactation persistency, highest for lactose yield) for hinds with records and 0.26–0.37 across all 834 genotyped animals, comparable to values reported for small dairy goat and sheep populations. Yield-trait GEBVs were strongly correlated (0.82–0.92); somatic cell score showed weak negative correlations with yields, and lactation persistency was weakly related to yield. Principal component analysis revealed only weak substructure among the four birth herds.

These first genomic breeding values for red deer milk traits provide a working evaluation for an emerging dairy species and demonstrate that GBS-based relationship estimation combined with cross-species parameter borrowing is a viable path for other small-population dairy evaluations.

Talk 26: Genetic Parameters for Fleece Shedding and Its Relationship with Growth Traits in New Zealand Sheep

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The New Zealand sheep industry is predominantly based on dual-purpose breeds; however, the declining profitability of coarse wool has increased interest in breeds that show natural fleece shedding (self-shedding). The aim of this study was to estimate the genetic parameters for fleece shedding and their relationships with key growth traits (including live weight and body condition score). Phenotypic data were collected across two farms and included shedding score (0-5 scale), and body condition score (1-5 scale) and live weight at birth, docking, weaning and pre-mating, with number of animals varying among traits according to data availability. Shedding was recorded at the lamb (~5 months) and hogget (~14-18 months) stages. Genetic parameters were estimated using genomic animal model. The heritability of lamb shedding score was estimated at 0.52 and hogget shedding score was 0.38, indicating a moderate to high level of genetic control and strong potential for genetic improvement through selection. Genetic correlation between lamb and hogget shedding scores was 0.79. Pearson correlations between shedding scores and production traits were low, while genetic correlation estimates varied among trait pairs. These findings support the potential of a self-shedding trait to be included into selection indices as a strategy for reducing shearing requirements while maintaining productivity in dual-purpose breeds.

Key words: Wiltshire; Romney; genetic parameters; scurs; body condition

Talk 27: Large-effect recessive variants impact sperm morphology and fertilisation traits in dairy cattle

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Male fertility is a complex trait underpinned by factors influencing sperm production, morphology, motility, and other functions. While the heritability of male fertility is reported to be low to moderate, contributions from rare, major-effect loci are also documented across species. Here, we performed both additive and non-additive genome-wide association studies using an array of sperm and reproductive success traits in dairy cattle. Using a dataset of ~6500 imputed or directly genome sequenced bulls, we highlight eight major quantitative trait loci (QTL), with most manifesting as recessive impacts on one or more traits. The allele frequencies of some of these recessive QTL are remarkably high, with homozygous affected animals together representing 5% of all bulls in the dataset. Retrospective analysis shows that homozygotes have been historically culled at much higher rates than contemporaries - accounting for 27% of all bulls culled for fertility-related issues. These QTL also present plausible missense and nonsense causative variants in most cases, mapping to genes encoding ciliary, spermatogenesis, and fertilisation factors. By examining bovine orthologues of a wider panel of curated human male infertility genes, we also show association enrichment for variants in these genes, proposing additional candidate causative variants below the significance threshold of our primary analyses.

Talk 28: How heritable is milk fatty acid composition in NZ dairy sheep?

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Milk fatty acid (FA) composition influences the nutritional profile, flavour, and processing properties of sheep milk. Genetic parameters for milk FA have been estimated mainly in Mediterranean dairy sheep breeds, with no estimates available for pasture-based New Zealand flocks. This estimated heritabilities of milk FA traits and genetic correlations among FA traits and with production traits in dairy sheep using Bayesian repeatability animal models, based on 624 milk samples from 173 ewes with pedigree records for 6,885 animals dating back to 1994. *De novo*-synthesized and odd- and branched chain FAs showed moderate heritability, suggesting potential for genetic improvement through selection. Correlations between protein percentage and FA traits were generally stronger than those between milk yield or fat percentage and FA traits, often in opposite directions. *De novo*-synthesized FA were generally negatively correlated with long-chain and unsaturated FA, potentially reflecting a competitive relationship between *de novo*-synthesis and uptake of preformed dietary FAs. The genetic correlations with production traits indicate potential to improve milk FA composition through selection while maintaining milk production performance. These findings provide a basis for incorporating milk FA composition into future breeding strategies aimed at improving sheep milk quality and processability.

Talk 29: Invited Speaker – Heidi Dungley
A story of genetics and trees

Speaker Biography

Heidi Dungey is a forest scientist working in the private sector with extensive experience in forest genetics, tree breeding, and the development of improved plantation forestry systems. Her work focuses on applying genetic innovation to enhance forest productivity, resilience, and sustainability. Heidi has led and contributed to numerous collaborative research and industry programmes and is passionate about translating scientific advances into practical outcomes for the forestry sector. Currently working for Stand Forestry, she is committed to developing innovative solutions that support the long-term productivity and resilience of planted forests.

Talk 30: Pangenome construction and targeted sequencing panel for apple breeding

David Chagné[1], Elena Lopèz-Girona[1], Sarah Bailey[1], Chris Kirk[1], Ignacio Carvajal[1], Elena Hilario[1], Melissa Taane[1], Linley Jesson[1], Astra Heywood[1], Ben Orcheski[1], Cecilia Deng[1]

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Comprehensively capturing genetic variation is critical for implementing genomic solutions for breeding in species that are important to New Zealand's primary production, such as perennial fruit trees. We are developing a pan-genome for 28 apple varieties, representing the New Zealand breeding germplasm. Near complete haplotyped assemblies are produced for these apples, with the goal of identifying single nucleotide and structural variants that would not be represented in a single reference genome. Genotyping solutions for screening breeding populations using a targeted sequencing method were developed to span this apple pan-genome. Building such genomic resources enables the discovery of variants associated with production traits such as disease resistance, and fruit quality traits such as appearance and eating quality. This allows us to identify causative variants for traits, including red flesh colouration. The development of such resources and knowledge supports genomic selection and germplasm characterization, ultimately accelerating the development of improved apple cultivars.

Talk 31: Microhaplotypes improve genomic predictive ability in white clover

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Microhaplotypes, short genomic loci comprising linked SNP variants within ~300bp, are inherited as multi-allelic haplotype markers. While exhibiting improved kinship analyses in several species, their value for genomic prediction (GP) in perennial forage crops remains largely unexplored. We evaluated the utility of microhaplotypes in allotetraploid allogamous white clover (*Trifolium repens*), a major perennial forage legume.

We investigated whether microhaplotypes could be identified and utilised from ~100bp genotyping-by-sequencing (GBS) tags. GBS data from 94 pooled full-sib lines in a white clover training population were used to detect microhaplotypes. Three marker datasets were generated for comparison in GP models: (i) SNPs underlying identified microhaplotypes (target SNPs); (ii) the full GBS SNP panel; and (iii) microhaplotype loci. After filtering, genomic relationship matrices were constructed using 1,169 target SNPs, 1,553 full SNPs, and 341 microhaplotype loci. Using these matrices, prediction modelling was conducted using rrBLUP with phenotypic data for Growth Score, Dry Matter, Stolon Number, Leaf Size, and a multi-trait selection index (Growth Score/Dry Matter.

Microhaplotypes consistently outperformed SNP-based markers. Predictive ability increased by 103% and 67% relative to target and full SNP panels, respectively, with improvements observed across all five traits. Standardized mean absolute marker effects were strongly correlated with predictive ability across all marker types and traits ($r > 0.98$, $p < 0.05$). Averaged across traits, microhaplotypes also exhibited larger mean absolute marker effects than either SNP panel.

These results demonstrate that microhaplotypes can substantially improve GS predictive ability in white clover and represent a promising marker system for forage breeding.

Talk 32: Multi-Omics Data Integration for the Discovery of Candidate Genes Involved in Dothistroma Needle Blight Resistance

Steffi Fritsche[1], Natalie Graham[1], Tancred Frickey[1], Emily McLay[1], Stuart Fraser[1], Joane Elleouet[1], Sai Arojju[2], Jules Freeman[3] & Jakob Butler[3]

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Dothistroma needle blight (DNB), caused by the fungal pathogen *Dothistroma septosporum*, poses a severe economic threat to New Zealand's plantation forestry, particularly for *Pinus radiata*, *Pinus attenuata,* and their hybrids. New Zealand's temperate, frequent-rainfall climate drives high DNB disease pressure, which can lead to severe defoliation and significant growth losses. Consequently, selection for increased DNB resistance is a key focus in radiata pine breeding and research programmes.

To support this goal, we integrated multi-omic data combining transcriptomic, genome-wide association study (GWAS), and quantitative trait loci (QTL) mapping of three independent data sets. Time-series RNA-seq profiling revealed key differentially expressed genes and a co-regulated network with a potential role in defence. Concurrently, GWAS identified putative associations with DNB resistance, while bi-parental QTL mapping from two *P. attenuata* × *P. radiata* crosses revealed 137 loci linked to DNB resistance. Finally, cross-referencing these independent datasets converged on 18 genomic regions in the current *P. radiata* genome assembly providing multiple lines of support for their role in DNB resistance and susceptibility.

Applying these multi-evidence DNB resistance and susceptibility regions to update genetic marker sets and prediction models will support an efficient and accurate selection for DNB resistance in tree breeding.

Talk 33: Beyond sequence: integrating structural data to reconstruct the evolution of bacteria movement

Caroline Puente-Lelievre[1,2], Jules Van Overbekend, and Nicholas Matzke[1,2]

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Bacteria swim by rotating flagella (tail-like organelles) while sensing chemical gradients and translating environmental signals into directed movement (chemotaxis). Both mechanisms are ancient and have diversified across the bacterial tree of life for billions of years. Uncovering how they arose is key to understand bacterial behaviour and design future pathogen control strategies.

We searched 620 bacterial genomes for homologues of two flagellar components: the stator proteins MotA and MotB, which power the flagellar motor, and CheY, the response regulator that switches its direction. We predicted the structure of each protein, translated it into the 3Di structural alphabet, which encodes the three-dimensional environment of each amino acid as a letter, and inferred phylogenies from amino acid and 3Di data.

Our findings suggest that the stator motor complex evolved once from a common ancestral ion transporter and acquired specialised structural traits to enable torque generation. In contrast, motility-related CheY proteins evolved at least twice from a wider family of response regulators. This large-scale comparative approach provides a powerful framework for investigating molecular evolution and highlights how protein structures capture the innovations that drive biological complexity and innovation.

Talk 34: Calling genomic variants at industrial scale

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_ Is the presenting author

Calling short variants is a well-understood problem for small cohort sizes. However, modern applications ranging from germline profiling to identifying somatic variants in medicine and agriculture require us to analyze entire eukaryotic populations. At this scale, the construction of an automatic pipeline faces significant challenges. We have solved these challenges over nearly a decade of developing population-scale bioinformatic infrastructure. In the process, we have built production pipelines capable of processing hundreds of individual genomes daily.

Here, we present the key lessons learned structured around three core architectural pillars:

- **Containerized Cloud Architecture:** Applying modular Nextflow workflows in cloud infrastructure to achieve reliable, rapid turnaround times.

- **Quality & Software Validation:** Integrating multi-layered sample controls to identify sample-level issues early, with version-controlled software releases to maintain analytical reproducibility.

- **Effective Data Storage & Delivery:** Combining relational databases and data products to streamline data retrieval and post-processing.

Automating variant calling at scale reduces costs by eliminating manual intervention, directly supporting outcomes in clinical diagnostics, disease management, and agricultural production.

Talk 35: GWAS of Facial Eczema (FE) in NZ dairy cattle

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Facial eczema (FE) is a potentially-fatal disease affecting cattle and other grazing ruminants such as sheep, caused by animals consuming pasture contaminated with the mycotoxin sporidesmin, produced by fungal spores. Facial eczema is characterised by photosensitivity causing skin irritation and peeling, as well as liver damage and lowered milk production. As FE symptoms are ultimately caused by liver damage, subclinical cases of FE are typically diagnosed by measuring levels of liver enzymes in blood, in particular gamma-glutamyltransferase (GGT).

Here, I present a GWAS analysis of GGT phenotypes from 18,000 cattle to detect loci affecting blood GGT levels, as a proxy measurement for FE tolerance. Using $\log_{10}$-GGT and binary GGT threshold phenotypes, several genome-wide significant additive and recessive quantitative trait loci (QTL) were discovered, with the most significant mapping to chromosomes 16, 17, 21, 23, and 26. Applying pathway analysis to these results highlighted pathways related to glutathione metabolism, iron and copper transport, and immune functions, with possible links to ferroptosis. These results may help improve the accuracy of genomic prediction for FE tolerance in dairy cattle.

Talk 36: From SNP Chips to Sequencing: Paths to Low-Pass and Targeted Genotyping for Livestock and Horticulture

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SNP arrays have driven genetic gain across Australasian agriculture for over a decade. They are standardised, robust and cheap per sample for established programs. But they only score the variants designed onto the chip, they are biased towards the breeds and cultivars used to design them, and they need a species-specific product. That makes them a poor fit for minor species, new traits, and the structural variation that increasingly shows up in breeding-value prediction.

Sequencing-based genotyping has now reached the point where those limits can be removed without giving up throughput or cost per sample. This talk compares two practical routes from arrays to NGS:

- Low-pass whole-genome sequencing with imputation: reads the whole genome at around 1× depth. With a suitable reference population, imputation accuracy above 0.95 is achievable, and the resulting breeding values match or improve on chip-based ones.

- Targeted capture sequencing: custom panels that carry over existing marker content and add new trait, parentage or pathogen targets. Panels can be changed without redesigning a chip.

Drawing on livestock and horticultural examples, we will cover:

- when each approach makes sense
- reference-panel requirements for imputation
- automation and miniaturisation strategies that make high-volume sequencing cost-competitive with chips
- a staged, low-risk path for running NGS alongside existing array programs before switching over

We aim to give breeders, testing labs and research groups a clear framework for deciding whether, when and how to make the move.

Talk 37: What affects genomic offset predictions? Insights from a simulation model of the common myna invasion

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Genomic offset (GO) is a tool that helps predict how individuals from a population will perform when moved to different environments. GO methods can therefore inform practical applications in agronomy, conservation, and invasive species management, particularly in the context of rapid global change. GO predictions rely on identifying gene-environment associations and then projecting fitness in a new environment based on the allele frequencies of these associated loci. However, these predictions rely on key assumptions, including that populations are locally adapted and that the clinal relationships between allele frequencies and environment extend beyond the observed environmental range used to train the model from. To address how GO performs in the context of numerous bottlenecks and population expansions, we used an individual-based model to simulate the common myna invasion from its native range in India into Australia, New Zealand, South Africa and the Pacific. GO was calculated using different permutations of training population sets (i.e., Australia-only, native-only, or all populations) and performance of these methods within Australia - representing the largest and most variable invasive populations - was evaluated by correlating fitness with GO. We found better GO performance for more locally adapted and genetically diverse source populations but also observed a complex pattern of performance across environmental space, particularly when projections extended past the training dataset's range. This highlights the difficulty in generalising patterns of GO performances with environmental novelty. Our results provide insights into variables affecting GO performance and a framework for species-specific simulations to help assess genetic methods.